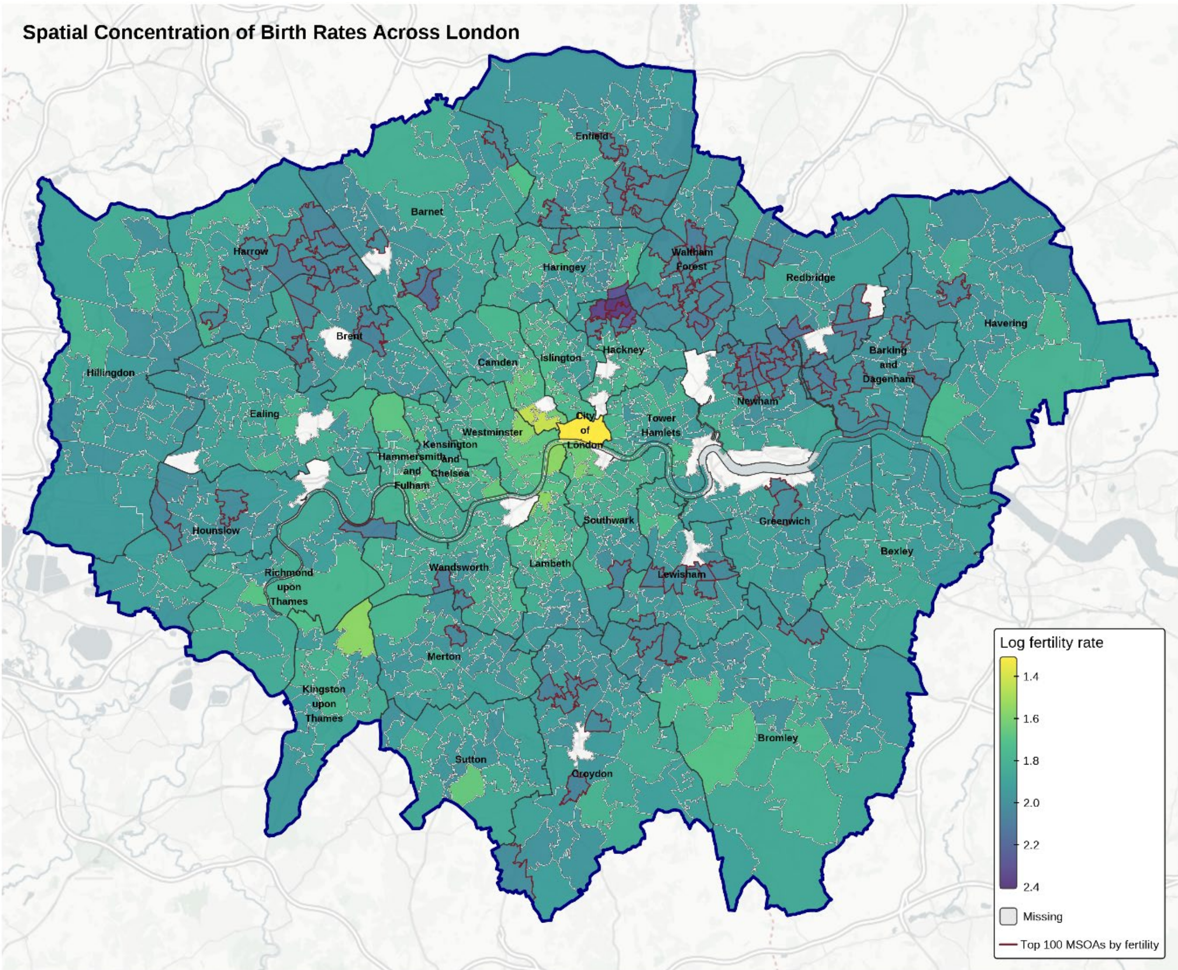
Where Families Still Form Under Housing Constraints in London

Family formation in high-density cities is often portrayed as a struggle, driven by rising housing costs and the dominance of small dwellings. In London, these pressures are particularly high, especially when fertility has fallen sharply to 1.35 children per woman in 2024 *(Births in England and Wales, 2024).*

Although declining birth rates are shaped by broader socio-demographic and economic factors, this analysis asks a distinctly urban question: **where in London does early family formation remain viable?**

As shown in the figure, fertility rates are relatively similar across areas outside Central London, yet distinct concentrations of higher fertility can still be identified, particularly in North and North-East London.



Drawing on neighbourhood-level data, this analysis examines how housing structure and cost shape the geography of young families across London. While high housing pressure is commonly assumed to push families outward towards the urban periphery, areas characterised by many small dwellings and high housing costs do tend to record lower birth rates, as noted in recent Assembly reports. These reports attribute a 20% decline in births since 2012 to an “unfavourable environment” shaped by housing and childcare costs *(London Assembly, 2025)*.

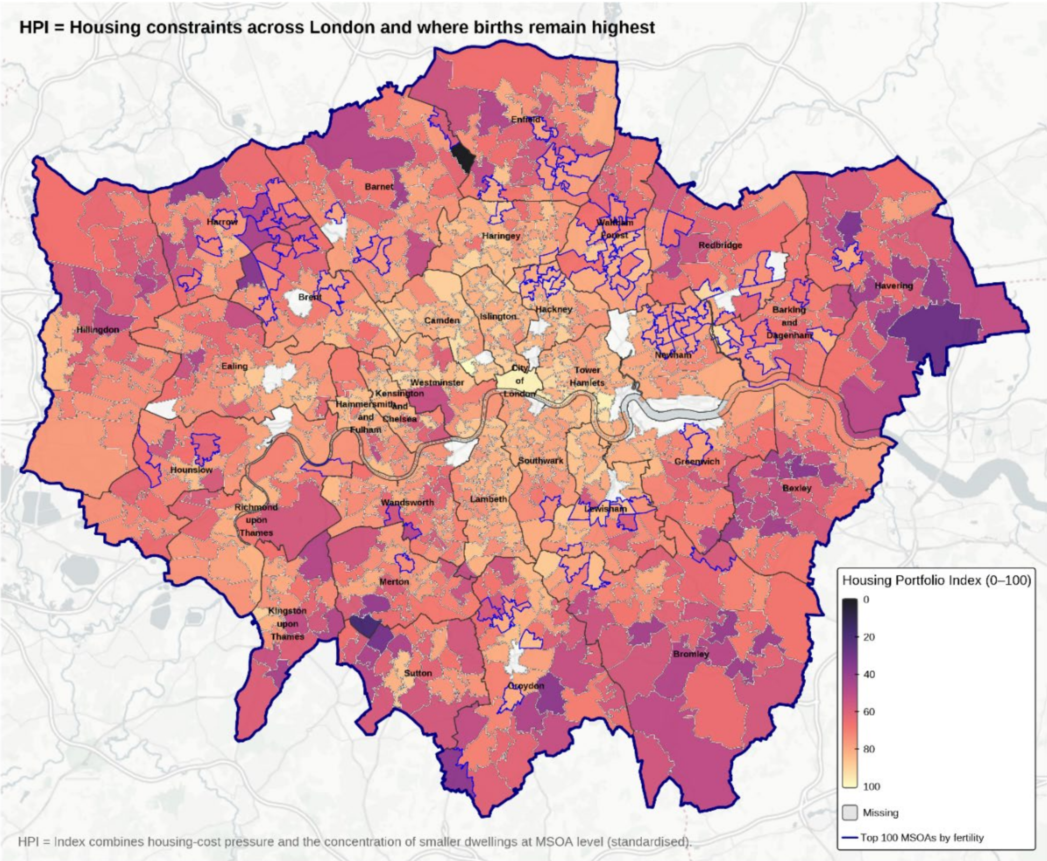
Yet this relationship is not universal. Some neighbourhoods maintain relatively high fertility despite intense housing pressure, suggesting that family formation can persist under conditions widely assumed to suppress it. At the city-wide scale, housing pressure emerges as a weak predictor of fertility; within borough contexts, however, it becomes more meaningful, revealing localised areas where families continue to form despite constraints.

Measuring Housing Pressure

To understand these patterns, the analysis focuses on two structural features of the housing environment: housing form and housing cost pressure. Housing form captures the size and type of dwellings; one-bedroom homes, while not a perfect proxy, signal a housing stock poorly aligned with family needs. The prevalence of these households shows which neighbourhoods have limited family space and which have more flexible households *(Dwellings by Property Build Period and Type, 2016).*

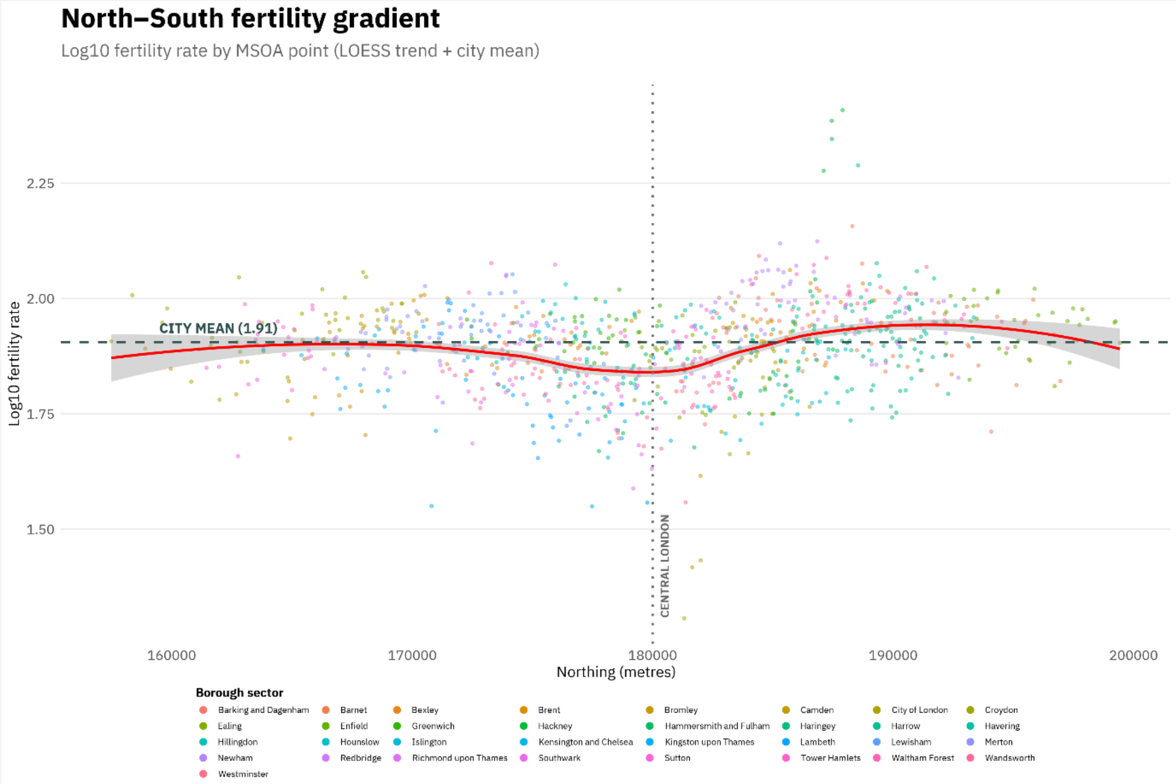
Housing cost pressure is defined as the gap between disposable income before and after housing costs, reflecting the financial burden imposed by the local market *(Income estimates, 2023)*. Combined, these measures form the **Housing Portfolio Index (HPI),** where higher values indicate neighbourhoods that are both expensive for the average household and structurally constrained.

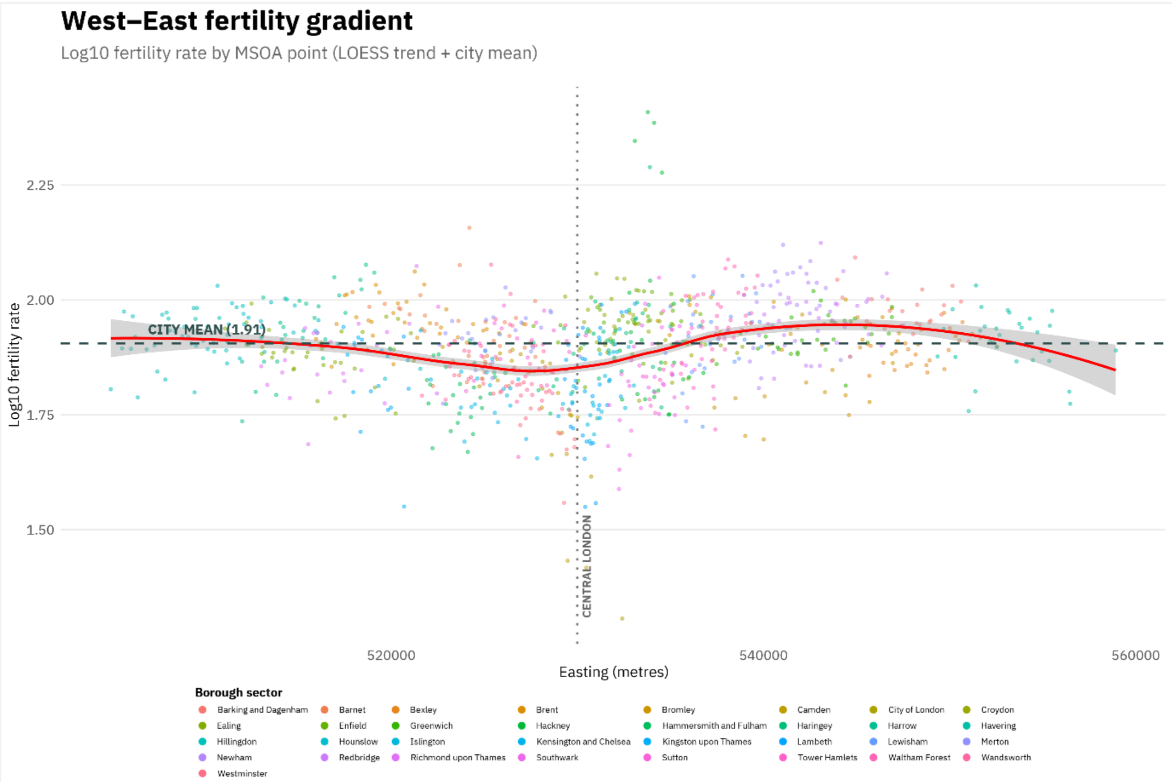
As shown in the figure below, HPI values are highest in Central London and generally decline with distance from the city center, suggesting lower housing pressure in outer areas. However, subsequent analysis demonstrates that housing pressure is more meaningfully understood within local and borough-level contexts rather than as a simple centre–periphery gradient.



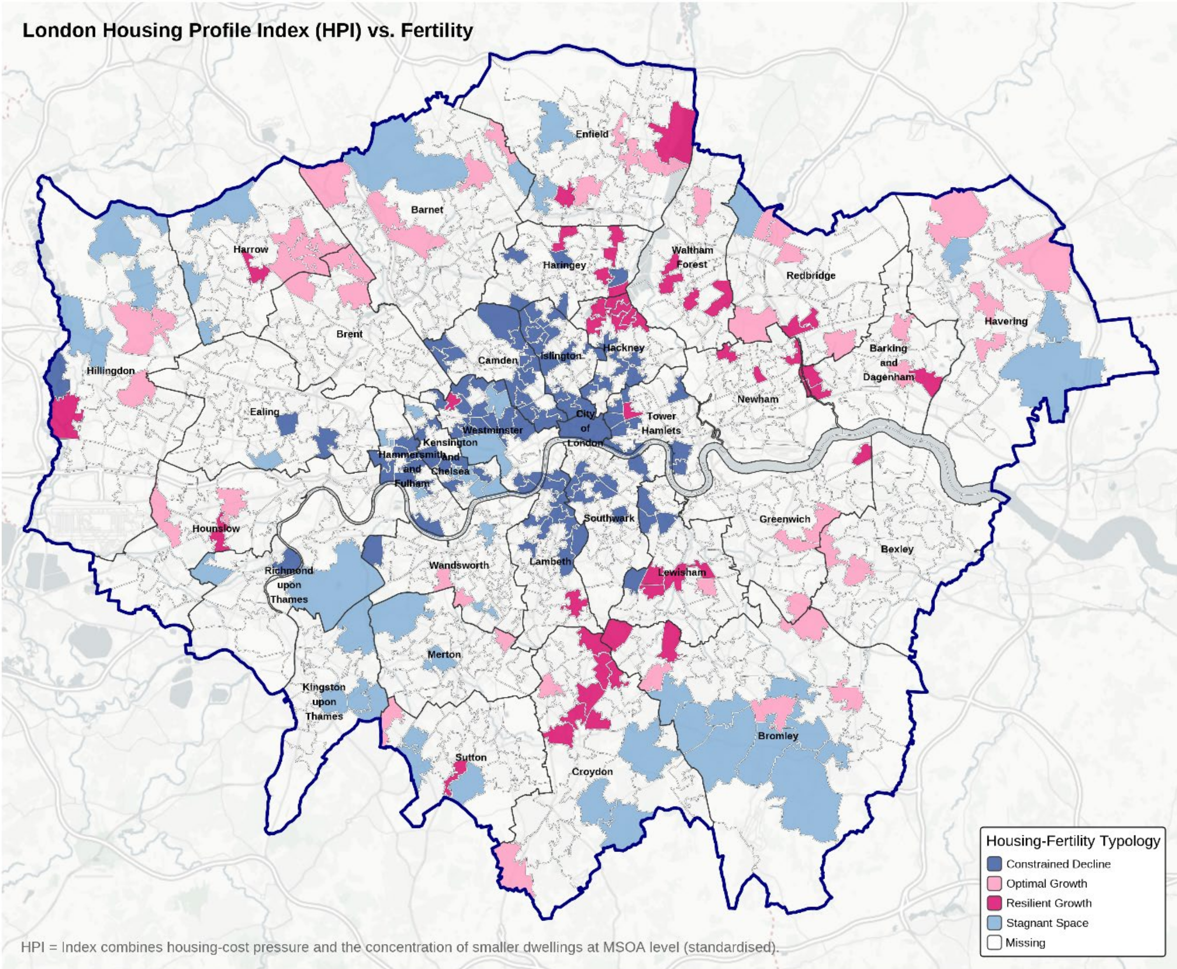
Fertility Across the City

Fertility rates *(Births and Fertility Rates, Borough 2021)* show a familiar pattern: lower rates in much of inner London and higher rates in outer boroughs. At first glance, at the north–south and west–east graphs, the gradients appear to support a localized outward family displacement, with fertility increasing away from the inner core towards the east and northeast zones.


North–South fertility gradient
Log10 fertility rate by MSOA point (LOESS trend + city mean)
CITY MEAN (1.91)
CENTRAL LONDON
Log10 fertility rate
2.25
2.00
1.75
1.50
160000
170000
180000
190000
200000
Northing (metres)
Borough sector
Barking and Dagenham
Barnet
Bexley
Brent
Bromley
Camden
City of London
Croydon
Ealing
Enfield
Greenwich
Hackney
Hammersmith and Fulham
Haringey
Harrow
Havering
Hillingdon
Hounslow
Islington
Kensington and Chelsea
Kingston upon Thames
Lambeth
Lewisham
Merton
Newham
Redbridge
Richmond upon Thames
Southwark
Sutton
Tower Hamlets
Waltham Forest
Wandsworth
Westminster


West–East fertility gradient
Log10 fertility rate by MSOA point (LOESS trend + city mean)
CITY MEAN (1.91)
CENTRAL LONDON
Log10 fertility rate
2.25
2.00
1.75
1.50
520000
540000
560000
Easting (metres)
Borough sector
Barking and Dagenham
Barnet
Bexley
Brent
Bromley
Camden
City of London
Croydon
Ealing
Enfield
Greenwich
Hackney
Hammersmith and Fulham
Haringey
Harrow
Havering
Hillingdon
Hounslow
Islington
Kensington and Chelsea
Kingston upon Thames
Lambeth
Lewisham
Merton
Newham
Redbridge
Richmond upon Thames
Southwark
Sutton
Tower Hamlets
Waltham Forest
Wandsworth
Westminster

Beyond Averages: Four Neighborhood Types



To understand the uneven relationship between housing pressure and fertility, London's neighborhoods can be grouped into four broad types:

1. **Constraint Decline:** High housing pressure and low fertility.
   Concentrated in the inner core, these neighborhoods align most closely with the conventional urban form of mixed-use areas with high concentrations of commercial and office space.
2. **Optimal Growth:** Low housing pressure and high fertility.
   Predominantly, outer-London areas where larger dwellings and lower costs coincide with higher birth rates and align with the narrative of inner migration.
3. **Stagnant Space:** Low housing pressure but persistently low fertility.
   Often, transitional or ageing neighborhoods are where fertility is shaped by factors beyond housing.

4. **Resilient Growth:** High housing pressure and high fertility.
   The most revealing category. These neighborhoods demonstrate that family formation can persist even under severe structural constraints.

Where Fertility Persists

Neighborhoods where families persist despite constraints cluster in East and North-East London, particularly Hackney, Haringey, Lewisham, Waltham Forest, and Newham. Here, high housing costs and many small dwellings coexist with fertility rates above the city average.

Hackney's example shows that this persistence is driven not only by housing, but by demographics and culture; a large Charedi Jewish population with high fertility, diverse ethnic groups with above-average birth rates, a young population, and ongoing inward migration all boost fertility *(City and Hackney Health and Wellbeing Profile, 2024).*

Importantly, this pattern becomes clearer within boroughs, where context matters. Early family formation operates at a local scale rather than as a purely city-wide phenomenon. Families are not uniformly displaced from high-cost, high-constraint areas; in some localities, they continue to form and maintain households despite these challenges.

Conclusion

The analysis identifies neighborhoods where family formation persists despite severe housing constraints and overlooked non-structural factors. These clusters challenge the assumptions of displacement. Rather than simply losing families to peripheries, London is highlighting where family life remains possible or rather, which areas are the last strongholds.

The findings suggest that future policies aimed at supporting early family formation must move beyond housing metrics alone. While affordability and dwelling size matter, borough-level context and non-structural factors play an equally significant role. These include demographic composition, cultural networks, access to schools and childcare, open space, and local amenities—factors that are primarily shaped by local councils rather than by city-wide master plans or housing supply strategies. As a result, many of the conditions that support family life remain underrepresented in conventional housing-focused policy frameworks, despite being central to neighbourhood viability for young families *(London Councils, 2025).*

Equally important are the dynamics of internal and international migration, which shape who enters, leaves, and remains in constrained neighbourhoods. Future research should incorporate these flows to better understand the conditions under which families choose to stay, settle, or relocate. Only by recognising the full ecology of family functionality—spatial, social, and economic—can urban policy develop a more nuanced and effective response to family life in high-cost cities.

Technical Appendix: Methodological Framework

Data, Methods, and Modelling Strategy

1. Data Sources and Preparation

This study integrates multiple open datasets at the Middle Super Output Area (MSOA) level to examine the relationship between housing conditions and fertility patterns across London.

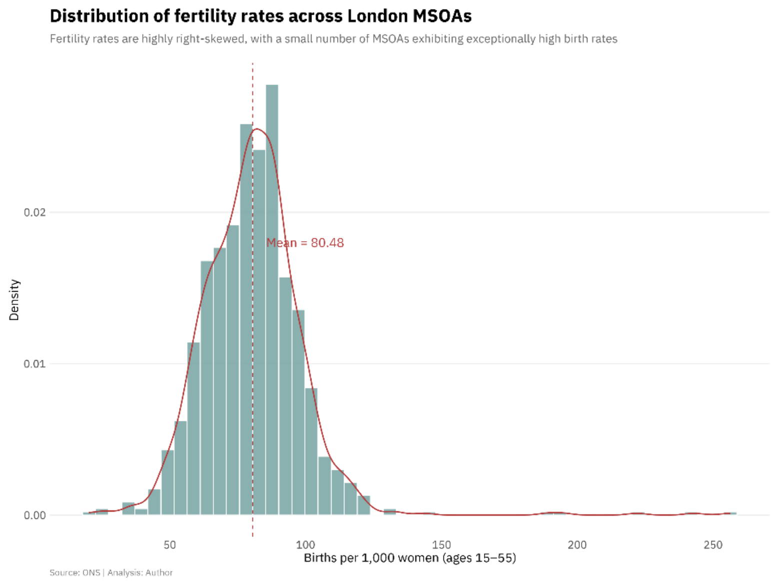
1.1 Births, Female population and Fertility

Birth data was obtained from the Greater London Authority's Births and Fertility Rates dataset. Birth records were aggregated over 2019–2021 to reduce annual inconsistency and smooth short-term fluctuations. Births were grouped by maternal age bands and summed for women aged 15–55, yielding cumulative births per MSOA over the three-year period *(Births and Fertility Rates, Borough 2021).*

Population denominators were derived from a female population dataset *(Population Statistics 2021)* filtered to the same age range (15–55) and aggregated to the MSOA level. Fertility rates were then calculated as:

$$\text{Fertility Rate} = \frac{\sum \text{Births}_{2019\text{–}2021}}{\sum \text{Female Population}_{15\text{–}55}} \times 1000$$

This produces a standardized measure of births per 1,000 women of childbearing age.



1.2 Housing Form: Dwelling Size Structure

Dwelling data were sourced from the London Dwellings by Property Type dataset (2016), reported at the MSOA level. Dwellings were grouped by number of bedrooms. To approximate constraints on family-sized housing, the analysis focused on the proportion of small dwellings (one-bedroom units) relative to larger ones (two-bedroom or more). Counts were converted into MSOA-level percentages *(Dwellings by Property Build Period and Type, 2016)*

1.3 Housing Costs

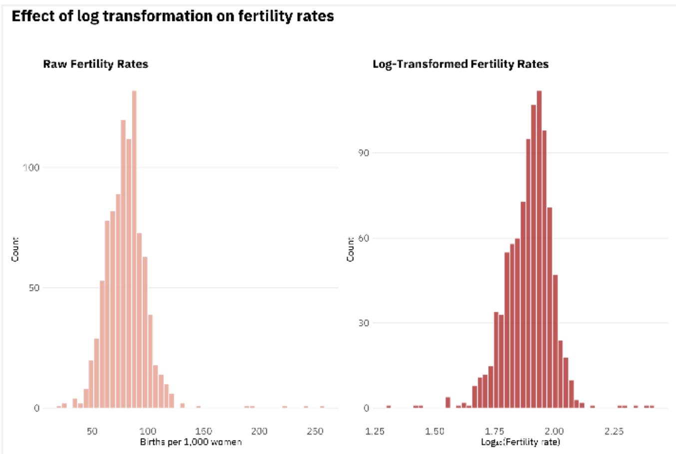
Income data were drawn from the Office for National Statistics Small Area Model-Based Income Estimates *(Income estimates, 2023).* Housing costs were operationalized as the difference between disposable (net) annual income before housing costs and disposable (net) annual income after housing costs. This measure captures financial pressure from housing costs per household but does not distinguish tenure type or household size.

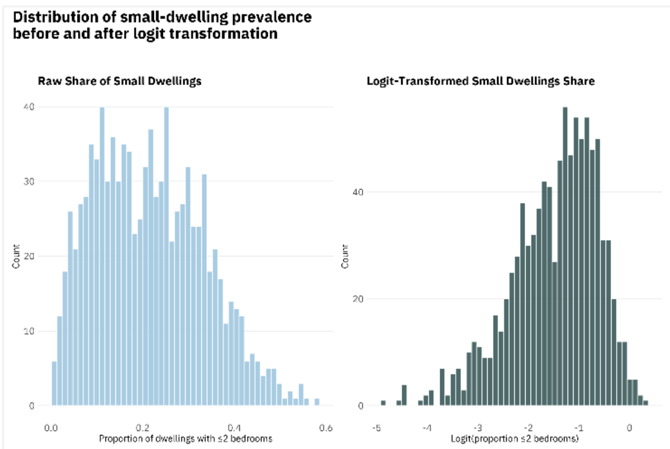
2. Variable Transformation and Correlation Structure

All continuous variables were transformed to improve normality and comparability:

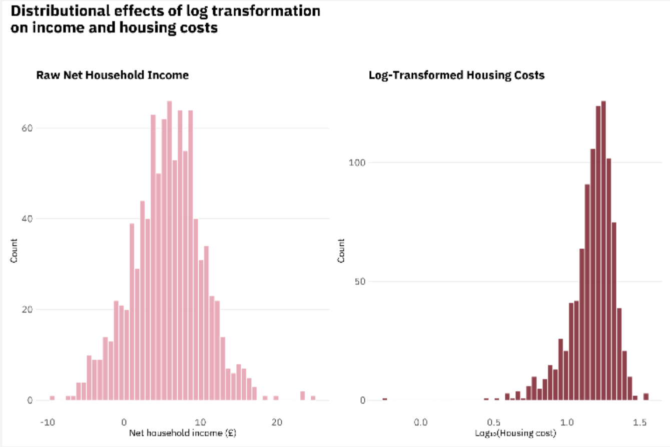
Fertility Rate ($\log_{10}$ transformed): To address the extreme positive skew and heavy right tail in the raw MSOA data.

Small Dwellings (Logit-transformed): Applied to the proportion of small dwellings to map bounded percentage data (0–1) onto an infinite scale.

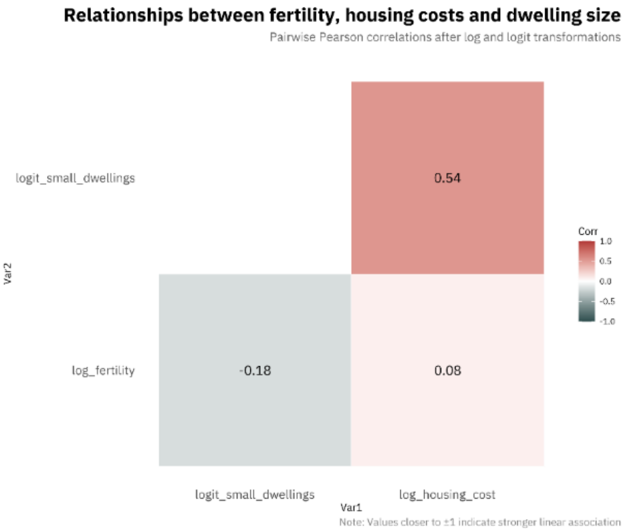




Housing Costs($\log_{\{10\}}$ transformed): Applied to stabilize the variance across the range of predictors, even where the raw income distribution appeared relatively symmetrically.



Pairwise Pearson correlations indicated a moderate positive correlation between housing costs and small-dwelling prevalence ($r = 0.54$), but only weak associations between individual housing variables and fertility. This motivated to approach it through PCA.



3. Construction of the Housing Portfolio Index (HPI)

To capture the multidimensional nature of housing pressure, Principal Component Analysis (PCA) was applied to the logit-transformed proportion of small dwellings and log-transformed housing costs. The first principal component (PC1) explained most of the shared variance, with both variables loading equally at 0.707.

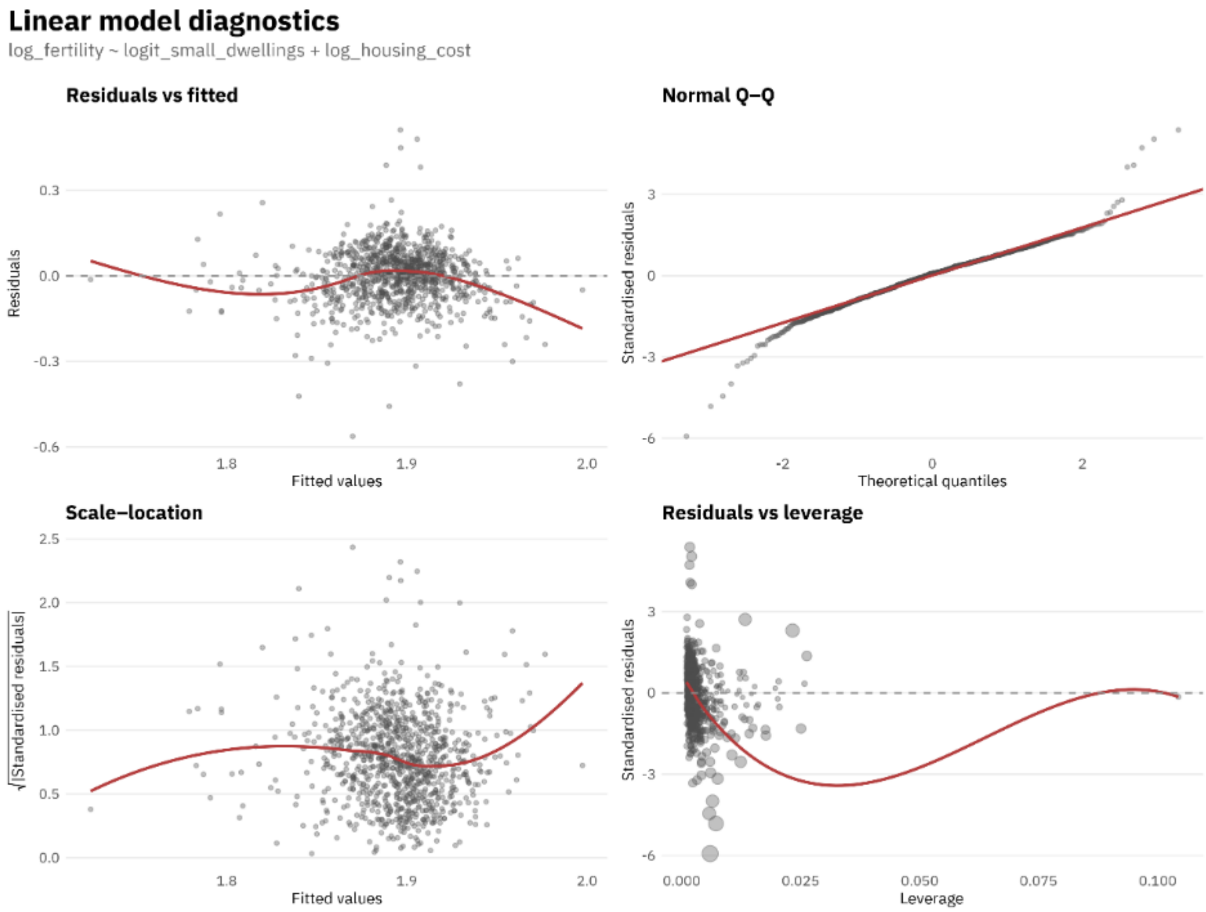
$$\text{PC1 (HPI)} = 0.707 \cdot \text{logit(small dwellings)} + 0.707 \cdot \log(\text{housing cost})$$

This component was retained as the **Housing Portfolio Index (HPI)**. Initial validation via a city-wide Pearson correlation revealed a weak negative association between the HPI and log-fertility ($r \approx -0.06$), suggesting that housing constraints do not exert a uniform linear influence across the capital, prompting to conduct an ANOVA test.

4. Inferential Modelling

4.1 Linear Regression

Baseline linear models showed that the negative association of small-dwelling prevalence (-0.031) was stronger than the positive association of housing cost (0.024). Despite statistical significance, explanatory power remained low ($R^2 \approx 0.07$). This suggests that the real drivers of fertility lie in unmeasured local factors, such as borough structure.



4.2 Borough-Level Effects (ANOVA)

A one-way ANOVA proved that birth rates vary drastically depending on which neighborhood is in ($F = 17.02, p < 2 \times 10^{\{-16\}}$). This high level of variation confirmed that a simple city-wide model was insufficient and required a more sophisticated mixed-effects approach that accounts for borough-level structures.

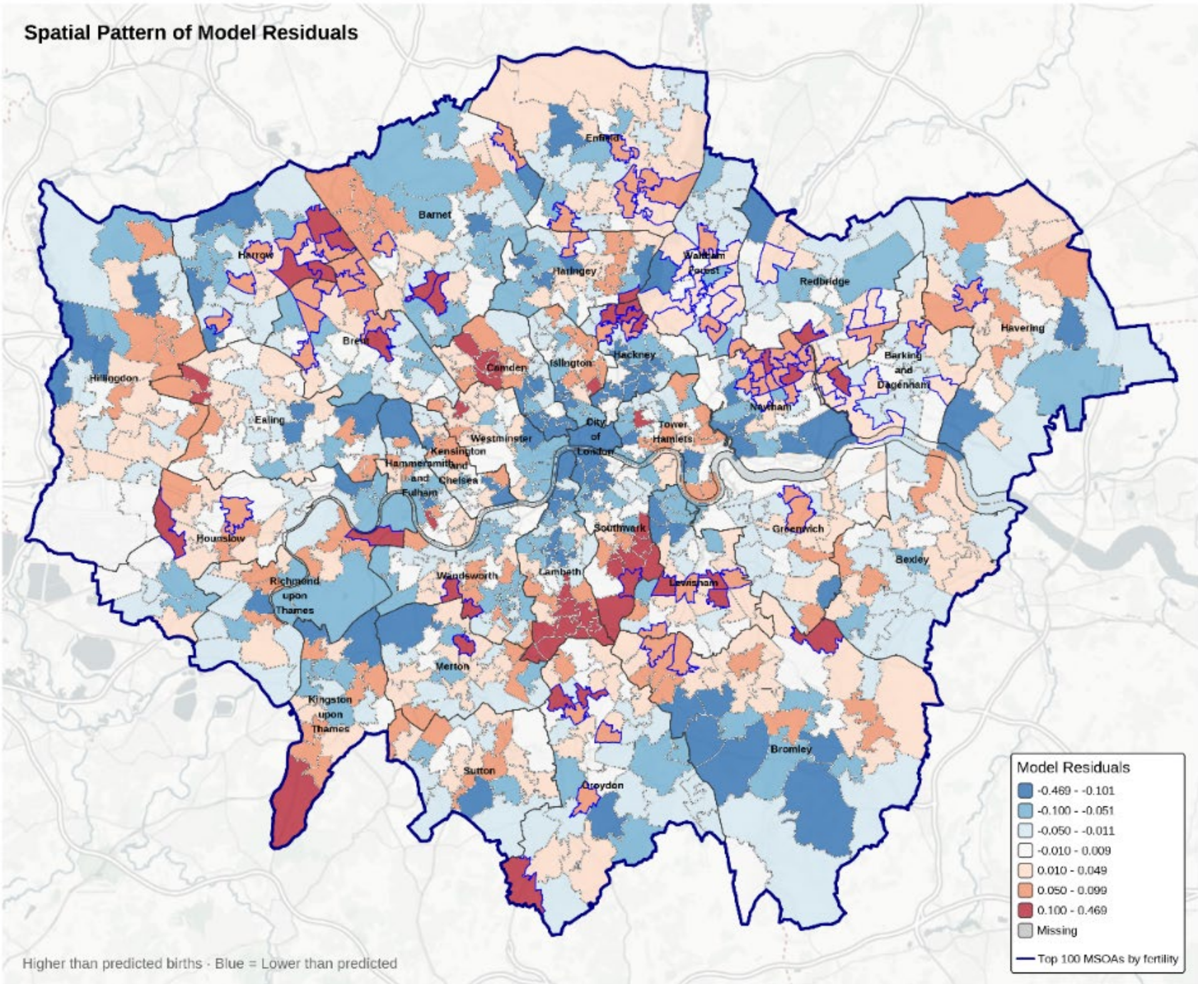
4.3 Linear Mixed-Effects (LME)

MSOAs were nested within boroughs to account for the hierarchical structure of London's administrative and social geography. The final Analysis was:

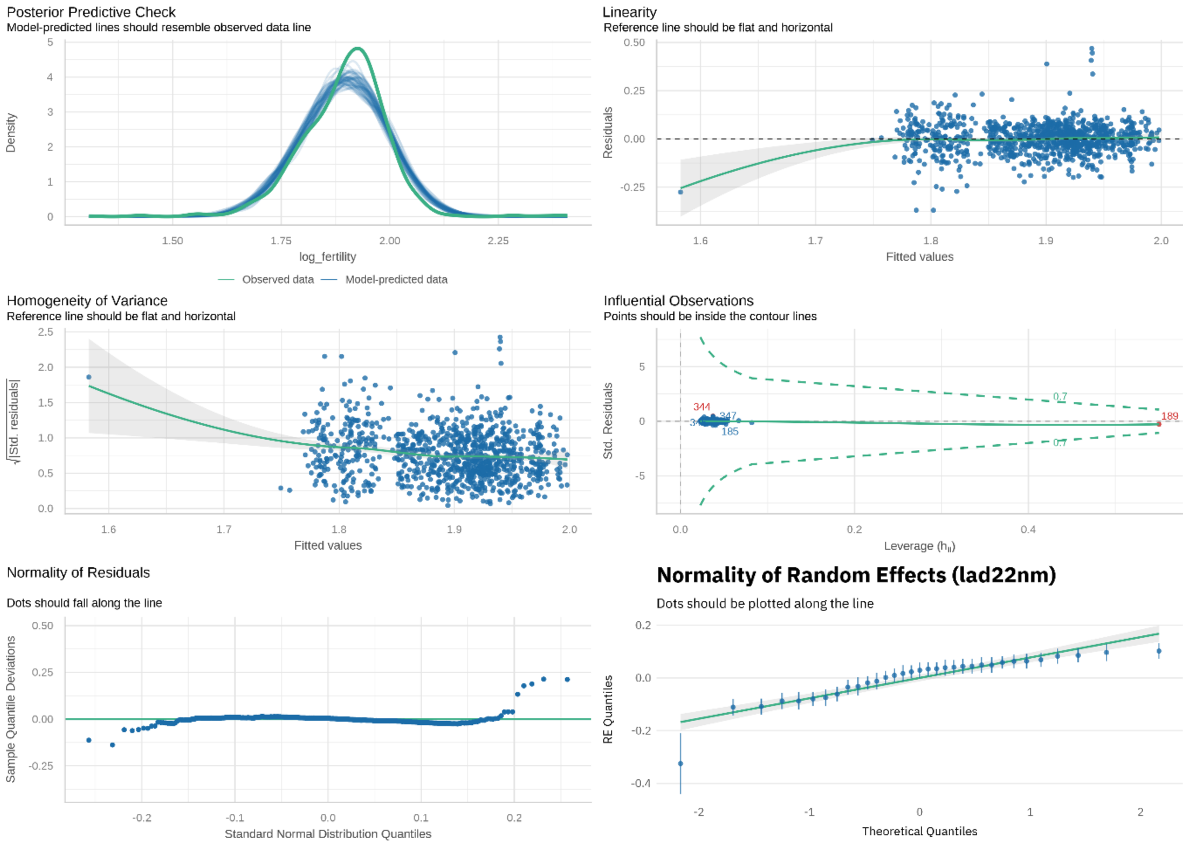
$$\log(\text{Fertility}) \sim \text{HPI} + (1 \mid \text{Borough})$$

The model confirms that the **Housing Portfolio Index** has a clear, statistically significant impact on birth rates ($p < 0.001$). However, the variation caused by **borough** structure (0.0075) is greater than that caused by individual neighborhood MSOA(0.0064). This means that the borough a family lives in and the social support it provides are just as critical to fertility as the physical size or cost of their home.

The residuals map below shows clear spatial clustering, indicating that fertility outcomes in some neighbourhoods remain higher or lower than predicted even after accounting for housing pressure and borough-level effects.



Diagnostics identified four high-zone MSOAs in Hackney, Camden, and the City of London. The Hackney outliers recorded the dataset's highest fertility rates ($\log_{\{10\}} > 2.38$), disproportionately affecting the borough's random intercept. These were retained to preserve the model's spatial integrity and represent genuine demographic "extreme zones". Robustness is confirmed by the Posterior Predictive Check, where the predicted data closely tracks the observed distributions, and by the Normality of Random Effects plot, which verifies hierarchical stability.

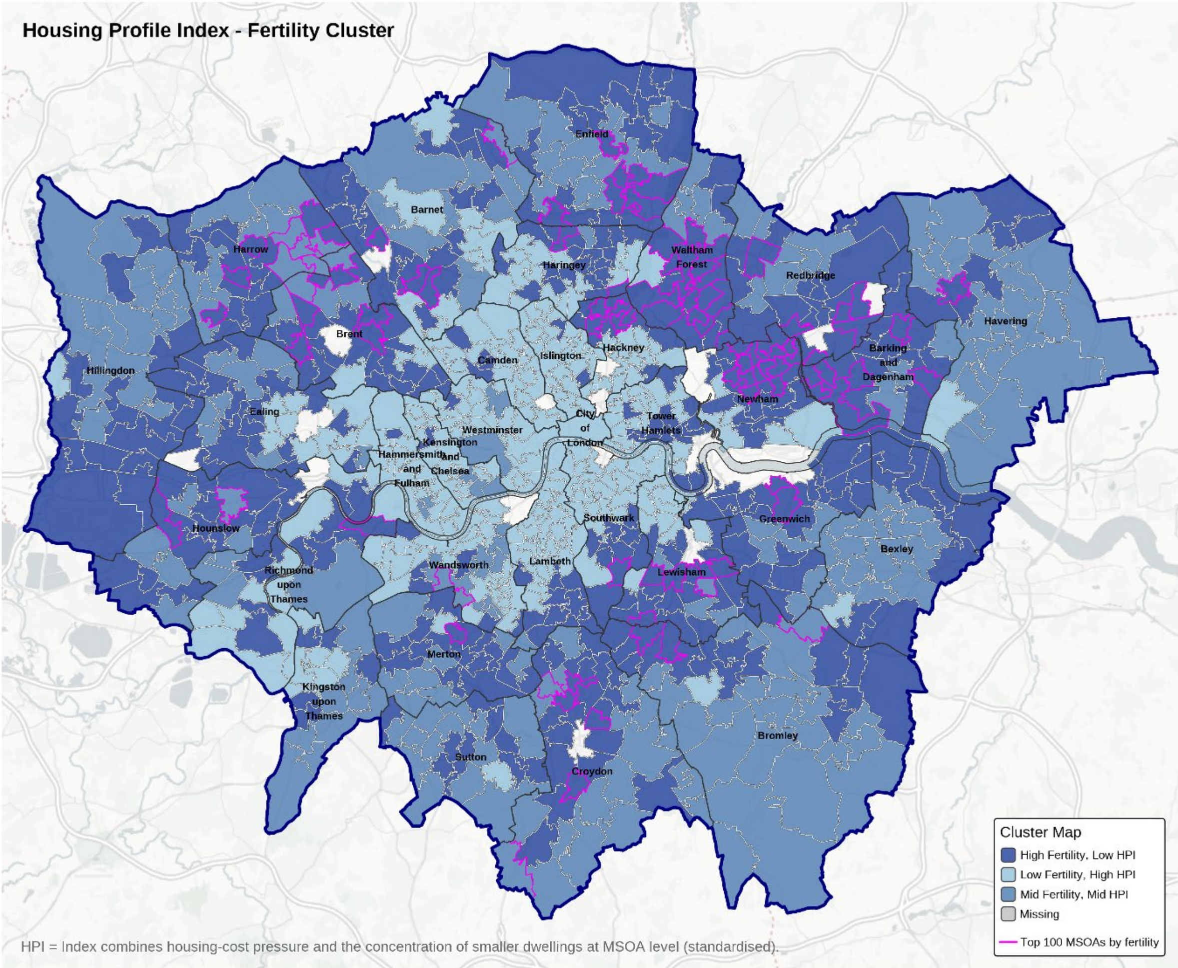


5. Cluster Analysis and Typology

K-means clustering was applied to standardized fertility, dwelling size, and dwelling cost data. A 3-cluster solution was selected based on the elbow method:

- **Cluster 1 (n=454):** Balanced baseline.
- **Cluster 2 (n=231):** High rooms, lower cost.
- **Cluster 3 (n=278):** Minimum rooms, highest cost areas.

The spatial distribution of clusters shows a clear center–periphery pattern, with high-pressure, low-fertility areas concentrated in inner London and lower-pressure, higher-fertility areas more common in the outer zones. Notably, several high-fertility MSOAs remain embedded within high-pressure clusters, reinforcing the presence of localised fertility existence under constraint.



Beyond broad clustering, a more selective typology map was needed to better understand London's existing structure.

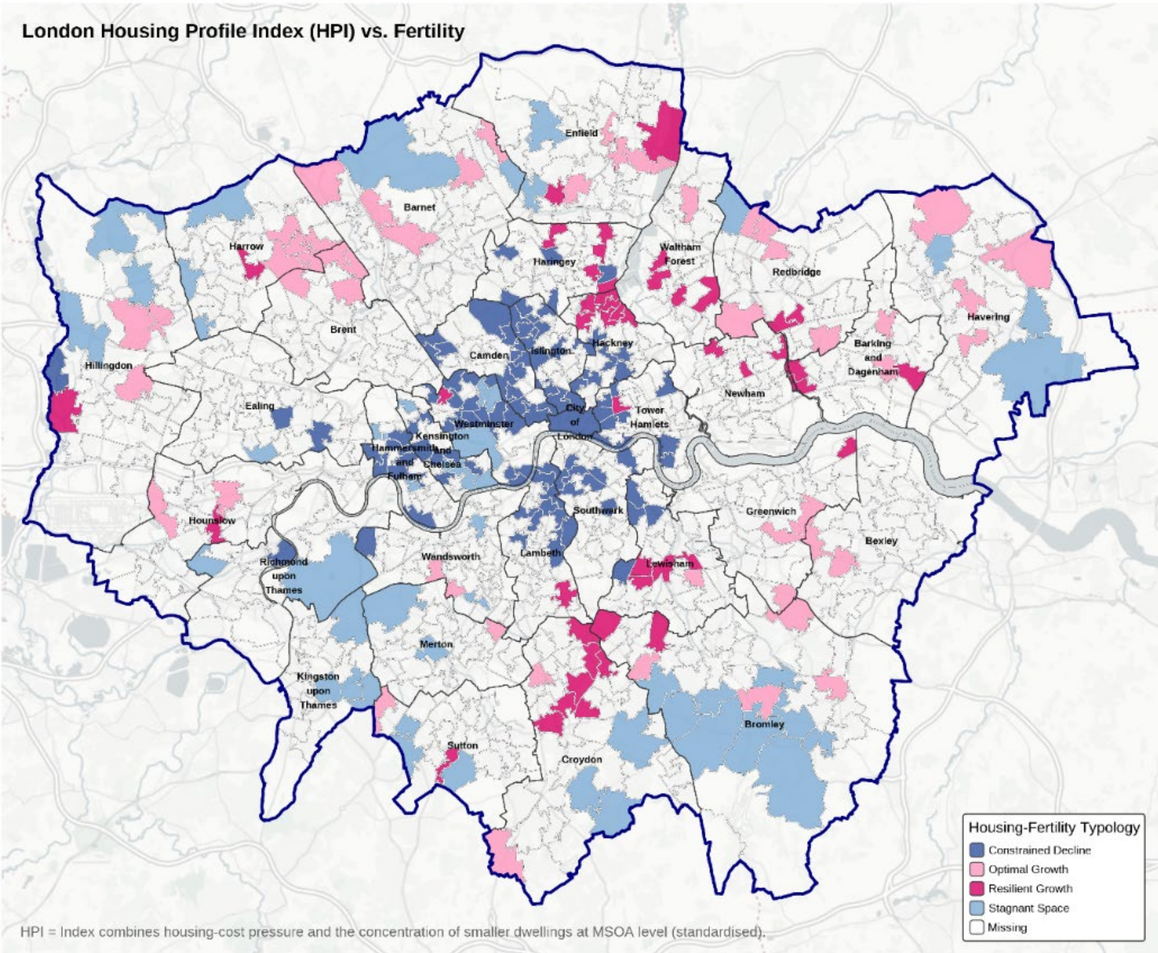
Categories are defined by quantile thresholds of fertility and the Housing Portfolio Index (HPI):

- **Resilient Growth (n=47):** High fertility (top 25%) despite high HPI constraints (top 25%); concentrated in **Hackney** (7), **Haringey** (5), and **Newham** (3).
- **Constrained Decline (n=103):** Low fertility (bottom 25%) coupled with high HPI constraints (top 25%); dominant in the **Inner-Urban Core**, specifically high-cost corridors of Central London.

- **Optimal Growth (n=50):** High fertility (top 25%) facilitated by low HPI constraints (bottom 25%); concentrated in **Periphery areas** such as **Croydon** (7), where larger housing portfolios support family formation.

- **Stagnant Space (n=49):** Low fertility (bottom 25%) despite low HPI constraints (bottom 25%); found in **Suburban pockets** like **Bromley** (2), where birth rates remain low despite more favorable housing conditions.

While housing constraints are statistically significant ($p = 0.000122$), fertility and early family formation persists, with localized social and structural factors mediating urban pressure.



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